UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

End of Term 1 Exam Coverage	هيكل اختبار نهاية الفصل الدراسي الأول
Grade 11 Advanced - Physics	**الصف الحادي عشر - متقدم - الفيزياء**
Academic Year 2026-2027	العام الدراسي 2027-2026
Inspire	**انسباير**

Number of questions/ عدد الأسئلة		Marks Per Question درجة السؤال	Maximum grade الدرجة القصوى
[1⇢16]	Number Of MCQ (16) عدد الاسئلة الموضوعية	2,3,4	100
[17⇢20]	Number of FRQs (4) عدد الاسئلة المقالية	9 ~11	

Exam duration مدة الاختبار	Mode of implementation طريقة التطبيق	Calculator الآلة الحاسبة
150 minutes/ 150دقيقة	Electronic & Paper / إلكتروني + ورقي	Allowed / مسموح بها

APPROVED

⚠ Questions might appear in a different order in the actual exam.

⚠ قد تظهر الأسئلة بترتيب مختلف في الاختبار الفعلي.

Q. T	Q.№		Learning outcome	Reference	Page
أسئلة اختيار من متعدد Multiple Choice Question	1		✓ Define a radian and convert degrees to radians and vice-versa.	S.T.B	3~4
	2		✓ Relate the linear velocity (v) to the angular velocity (ω) and the distance (r) from the axis of rotation.	S.T.B	5
	3		✓ Relate the linear acceleration (a) to the angular acceleration (α) and the distance (r) from the axis of rotation.	S.T.B / P.P(3,4)	5 , 6
	4		✓ Define angular frequency and relate the frequency (f) to angular frequency (ω). ✓ Solve problems related to rotational variables.	S.T.B P.P(2) / C.Y.P(6,8,10)	6 6
	5		✓ Show that different points of a rotating rigid object (Earth is an example) do not move the same distance but rotate through the same angle. ✓ Explain that different points of a rigid object rotate at the same rate	S.T.B	4~5
	6		✓ Identify that a torque is assigned a positive or negative sign depending on the direction it tends to make the body rotate about a specified rotation axis: “clocks are negative”. ✓ Calculate the net torque when more than one torque acts on a body about a rotation axis.	S.T.B E.P(1) P.P(11,12,13,14)	8 9 9
	7		✓ Calculate the moment of inertia (rotational inertia) of a single point mass rotating about an axis ($I=mr^2$) . ✓ Find the total moment of inertia due to many point masses	S.T.B E.P(3) / P.P(21,24)	12 14 , 15
	8		✓ Locate the center of mass of an object experimentally by suspending it from different points. ✓ Explain how the location of the center of mass affects the stability of an object.	S.T.B C.Y.P(42,44,45)	18~20 24
	9		✓ State the conditions for an object to be in static equilibrium- both translational and rotational equilibrium.	S.T.B / C.Y.P(41)	20 , 24
	10		✓ Describe the Coriolis force as an apparent force in a rotating frame of reference, which appears to deflect an object from its path.	S.T.B	23~24
	11		✓ Explain what happens when a player hits a ball in terms of the acting forces and their effects on the colliding objects before, during, and after collision. ✓ Derive impulse expression from Newton's Second Law of motion and specify its unit.	S.T.B	30~31
	12		✓ Calculate the impulse in case of a constant force and in case when the force is not constant by using the average force multiplied by the time interval over which it acts or by finding the area under a force-time graph.	S.T.B P.P(3)	30~31 33
	13		✓ Conduct experiments or simulations that demonstrate change in velocities and momentum during collisions. ✓ Define the system and apply the law of conservation of linear momentum to different situations in one dimension like recoil, propulsion in space, or others to calculate different physical quantities.	S.T.B E.P (3,4) P.P(17,18,20) / P.P(21,22)	38~42 40,42 40, 43
	14		✓ Investigate some applications of the impulse-momentum theorem in everyday life. ✓ Explain how impulse-momentum theorem is used to save lives in case of car air bags.	S.T.B C.Y.P(11)	33 37
	15		✓ Explain that an object's angular velocity can change if its shape changes even if no torques act on it and its angular momentum is constant. ✓ Discuss some examples where the angular impulse-angular momentum theorem is involved, like diving into a pool or during ice-skating.	S.T.B C.Y.P(15)	34~35 37
	16		✓ Define the system and apply the law of conservation of angular momentum to solve relevant problems. ✓ Discuss different examples that involve the law of conservation of angular momentum such as ice skating, tops, gyroscopes, and others.	S.T.B Get it? C.Y.P(28)	45~47 46 47
أسئلة مقالية Free Response Questions	17	Part 1	✓ Define angular displacement. ✓ Apply the relationship between average angular velocity, angular displacement, and the time interval for that displacement.	S.T.B P.P(1,2) C.Y.P(7,9)	4~5 6 6
		Part 2	✓ Apply the relation ($\tau=Frsin\theta$) to calculate the magnitude of torque (τ) for a force of magnitude (F) where (r) is the distance from the axis of rotation to the point where the force is exerted, and (θ) is the angle between the force and the radius from the axis of rotation to the point where the force is applied. ✓ Solve problems that involve torque. ✓ Solve problems related to objects in equilibrium.	E.P(1)/E.P(2)/E.P(5) P.P(18) P.P(38,39,40)	9,11,21 12 22
	18	Part 1	✓ Apply newton’s second law for rotation to solve problems. ✓ Determine the moment of inertia of extended objects like the hoop, solid uniform cylinder, uniform sphere, long uniform rod, and rectangular plate	S.T.B P.P(23) / E.P(4) P.P(30) / C.Y.P (33,34,35)	13~15 15,16 17
		Part 2	✓ Define the angular momentum of an object as the product of the object's moment of inertia and the object's angular velocity, and write it in equation form $L= I\omega$ and specify its unit. ✓ State the angular impulse-angular momentum theorem and write it in equation form $\tau\Delta t = L_f - L_i$ ✓ Solve problems involving angular momentum.	S.T.B E.P(2) / P.P(6,7)	34 36,37
	19		✓ Apply the linear momentum equation to calculate the linear momentum of different systems. ✓ Apply the impulse-momentum theorem to solve relevant problems. ✓ Solve problems involving impulse.	E.P(1) P.P(1,2,4) C.Y.P (12)	32 33 37
	20		✓ Explain the law of conservation of linear momentum and express it in equation form. ✓ Define the system and apply the law of Conservation of linear momentum to different situations in one dimension, like recoil, propulsion in space, or others, to calculate different physical quantities. ✓ Apply the impulse-momentum theorem to solve relevant problems	S.T.B E.P (3,4) P.P(17,18,20) / P.P(21,22)	38~43 40,42 40, 43

Q.T: Question Type	Q.№: Question Number	S.T.B: Student Book	P.P: Practice Problems.
Reference: Reference(s) in the Student Book (English Version or Arabic Version).		E.P: Example Problem.	C.Y.P: Check Your Progress.